

Exercise Questions: Decision Trees

These exercises use small, simple datasets so that the calculations can be done by hand.

Exercise 1 — Choose the Best Attribute

A school wants to predict whether a student will Pass based on Study Hours and Attendance.

Use the dataset below. Calculate the entropy of the target variable Pass and the information gain for Study Hours and Attendance. Which attribute should be the root node?

Study Hours	Attendance	Pass
Low	Low	No
Low	High	No
High	Low	Yes
High	High	Yes
High	High	Yes
Low	High	No

Exercise 2 — Build a Small Decision Tree

A shop wants to predict whether a customer will Buy a product based on Age Group and Advertisement.

Construct a decision tree using information gain. You only need to find the root node and the branches immediately below it.

Age Group	Advertisement	Buy
Young	Yes	Yes
Young	No	No
Adult	Yes	Yes
Adult	No	No
Senior	Yes	No
Senior	No	No

Exercise 3 — Calculate Gini Impurity

For the following dataset, calculate the Gini impurity of the target variable Weather. Then calculate the weighted Gini impurity after splitting by Temperature.

Temperature	Play
Hot	No
Hot	No
Mild	Yes
Mild	Yes
Cool	Yes
Cool	No

Exercise 4 — Prediction Using a Given Tree

Suppose a decision tree uses Study Hours as the first split:

Study Hours = Low → Fail

Study Hours = High → Pass

Use this tree to predict the result for the following students:

Student	Study Hours
A	Low
B	High
C	Low
D	High

Exercise 5 — Identify Overfitting

A decision tree gives 100% accuracy on the training data but only 70% accuracy on new test data.

a) Is this a possible sign of overfitting? b) Give two simple methods that can be used to reduce overfitting in a decision tree.